SemanticQA

Revisiting *a Pain in the Neck*: A Semantic Reasoning Benchmark for Language Models

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Motivation: A Blind Spot in LM Evaluation

Existing benchmarks emphasize math, code, and logic, but overlook **phrasal semantics**:

- **Sub-sentential meaning:** where meaning emerges from the interplay of lexical components and context.
- **Heuristic shortcuts:** even frontier LMs lean on surface heuristics rather than stable phrase-level representations.
- **Conflated formats:** existing tasks bundle multiple semantic operations, so strong scores may not reflect understanding.

We need diagnostic evaluation that ***disentangles*** *semantic operations.*

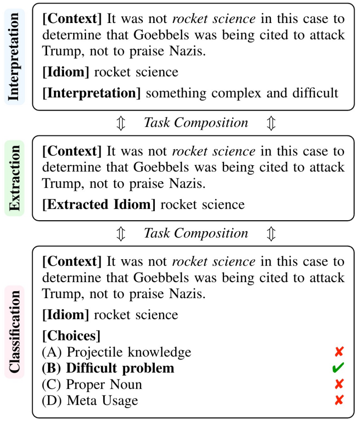

Figure 1: Atomic task exemplars of idiomatic expression, grouped as task compositions.

Research Question & Contributions

*How do language models behave when evaluated on phrasal semantics across **distinct** but **structurally constrained** task operations?*

SemanticQA contributes on three fronts:

1. **Operation-aligned Semantic Evaluation:** aligns existing SP tasks with the *semantic operations* they instantiate, enabling cross-operation analysis on the same phrase meaning.
2. **Minimal and Controlled Design:** fixed prompt templates hold structure constant while varying operations, supporting fair comparison under shared conditions.
3. **Diagnostic Analyses of Cascade Sensitivity:** sequential task compositions reveal phrase-level limitations that remain hidden in single-task evaluations.

SemanticQA: Framework Overview

SEMANTICQA benchmarks LMs on **4 phrase types × 3 atomic operations**:

- **Phrase types:** Idiomatic Expressions (IE), Lexical Collocations (LC), Noun Compounds (NC), Verbal Constructions (VC).
- **Atomic operations:** *Classification* (multi-choice), *Extraction* (span identification), *Interpretation* (paraphrase generation).

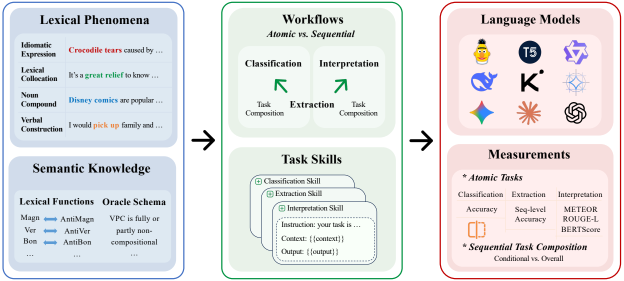

Figure 2: Overview of SEMANTICQA for benchmarking LMs on lexical phenomena.

Phrase Type Taxonomy & Coverage

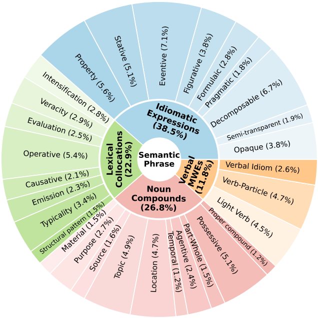

Figure 3: Coarse- and fine-grain semantic phrase categories of SEMANTICQA.

4 phrase types × 3 atomic operations ⇒ ***12 task variants****, spanning the full compositional spectrum.*

Four representative phrase types capture popular sources of variation:

- **LC:** *base + collocate* (*e.g.*, *heavy rain*); compositional to idiom-style.
- **IE:** non-compositional (*e.g.*, *kick the bucket*); needs conventionalized meaning.
- **NC:** compositional but ambiguous (*e.g.*, *baby oil vs. olive oil*).
- **VC / VMwE:** VPC, LVC, VID; semi-compositional verbal expressions.

Together, they expose ***distinct*** *sources of semantic difficulty.*

Three Atomic Operations on the Same Phrase

Each phrase meaning is probed by **three structurally distinct operations** sharing a common context $\mathcal{S}$:

- **Extraction:** identify the target span in $\mathcal{S}$ (strict structural constraint).
- **Classification:** pick the correct sense from candidate options.
- **Interpretation:** generate a contextualized paraphrase (open-ended).

Context: "*It was not rocket science in this case to determine that Goebbels was being cited…*"
Extraction: *rocket science* ✓
Classification: (A) Projectile knowledge ✗ **(B) Difficult problem** ✓ (C) Proper noun ✗
Interpretation: "something complex and difficult" ✓

Benchmark Construction & Datasets

SEMANTICQA unifies prior resources into a single **operation-aligned** testbed:

- **IE:** detection, extraction, interpretation (273 / 447 / 818 examples).
- **LC:** categorization, extraction, interpretation (305 each, balanced by relation).
- **NC:** compositionality, extraction, interpretation (242 / 720 / 110).
- **VMwE:** VPC, LVC, VID extraction (475 examples total).

Phrase Type	Tasks	Output Constraint
Idiomatic Expressions	IED, IEE, IEI	Choice / Span / Paraphrase
Lexical Collocations	LCC, LCE, LCI	Choice / Span / Paraphrase
Noun Compounds	NCC, NCE, NCI	Choice / Span / Paraphrase
Verbal MwE	VPE, LVE, VIE	Span

Table 1: SEMANTICQA tasks. Datasets are deduplicated and reformatted for consistency across operations.

Overall Performance: The Capacity Triangle

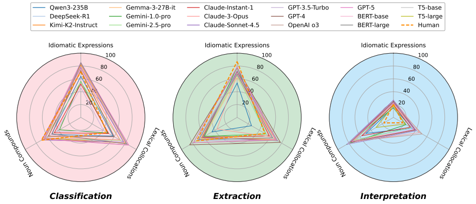

Figure 4: Best-performance capacity triangle (△) of LMs on SEMANTICQA.

Axes: Extraction (Acc_S) | Classification (Acc) | Interpretation (MTR); each axis aggregates over 4 phrase types.

Each axis is one atomic operation. A balanced model fills the triangle:

- **No model dominates uniformly:** strengths vary by operation *and* phrase type.
- **Operation-specific:** each operation imposes distinct structural constraints.
- **Diagnostic, not a leaderboard:** SEMANTICQA is neither saturated nor uniformly difficult.

*Triangle area indexes **breadth**; vertices reveal **operation-specific** weak spots.*

Main Results: LMs Struggle Differently Across Operations

Model	Idiom (IE)			Collocation (LC)			Noun Comp. (NC)			VMwE
	IED	IEE	IEI	LCC	LCE	LCI	NCC	NCE	NCI	avg.
HUMAN	71.0	**87.0**	20.5	47.0	50.0	16.7	**71.0**	73.0	17.2	**72.7**
DEEPSEEK-R1 (5-shot)	84.3	72.3	19.2	76.1	64.3	32.9	60.6	70.7	68.7	58.2
KIMI-K2-INST. (5-shot)	81.7	69.6	21.7	79.7	69.2	36.9	64.7	63.6	76.7	57.0
CLAUDE-SONNET-4.5 (5-shot)	78.0	72.0	**26.7**	76.1	**72.7**	40.8	70.1	62.1	**83.8**	55.6
OPENAI O3 (5-shot)	83.5	74.7	21.9	83.6	71.5	35.9	63.5	78.6	74.5	54.7
GPT-5 (5-shot)	**85.4**	**78.7**	22.5	84.3	68.9	37.4	67.2	**79.0**	75.3	54.5

Table 2: Major results. Acc / Acc_5 / MTR depending on task. Even frontier LMs trail humans on extraction and interpretation.

- **No single winner:** different LMs lead on different operations and phrase types.
- **Persistent gap on interpretation:** MTR remains low *(e.g., IEI < 30%)* despite strong B-S, indicating fluency outpaces grounding.

Cross-Task Variance: Mean vs. Std by Category

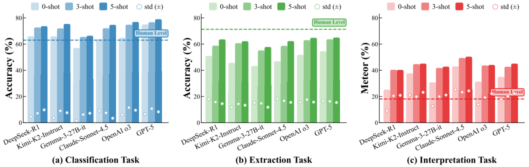

Figure 5: Bars = mean per category; markers = population std across tasks.

Read this as: bar height = *average* score; marker spread = *volatility* across the 12 tasks. Tall + tight ⟹ broadly competent; tall + wide ⟹ uneven.

What the variance reveals:

- **High std** on extraction signals *operation-specific* brittleness, not uniform difficulty.
- **Interpretation:** compressed mean but moderate std under flexible metrics.
- Strong mean ≠ consistent competence; **cross-task spread matters**.

*Mean reports **average** skill; std reports **stability** — both are needed.*

Semantic Category Scaling with In-Context Learning

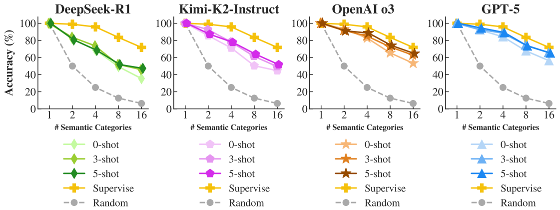

Figure 6: LC categorization with $n \in \{1, 2, 4, 8, 16\}$ classes and $k \in \{0, 3, 5\}$ shots.

> *Larger n exposes a steeper decay; more shots help only marginally.*

Findings on category scaling:

- All LMs beat random / majority baselines $\Rightarrow$ non-trivial semantic reasoning even zero-shot.
- Accuracy **drops sharply** as n grows: DeepSeek-R1 81.7% $\rightarrow$ 35.4%.
- Supervised baselines degrade more gracefully than frontier LMs.

> *ICL alone does not substitute for explicit supervision under fine-grained relational distinctions.*

Sequential Task Compositions: Where Pipelines Break

Realistic workflows chain operations: *extract* $\longrightarrow$ *interpret* or *extract* $\longrightarrow$ *classify*.

- **Cond. MTR** (on correctly-extracted phrases) is consistently **much higher** than **Overall MTR** (end-to-end).
- Interpretation *cannot rescue* upstream extraction errors $\Longrightarrow$ structural errors propagate.
- Compositional classification degrades sharply as relational granularity grows (4 $\longrightarrow$ 8 $\longrightarrow$ 16 classes).

Type	Model	Ext. (Acc)	Cond. (MTR)	Overall (MTR)
LC	DeepSeek-R1 5-shot	33.8	**42.3**	14.3
	GPT-5 5-shot	41.3	41.8	**17.3**
IE	DeepSeek-R1 5-shot	57.0	13.4	7.6
	GPT-5 5-shot	59.3	**17.1**	**10.1**

Table 3: Sequential extraction $\longrightarrow$ interpretation. Cond. MTR $\gg$ Overall MTR exposes the extraction bottleneck.

VMwE Extraction with Oracle Schema

Oracle Schema: augment the prompt with the *target VMwE type* and its definition (VPC / LVC / VID).

- Consistently lifts performance across frontier LMs.
- DeepSeek-R1: **51.6%** $\longrightarrow$ **64.1%** on VMwE extraction.
- Explicit semantic descriptions of the target structure substantially aid identification.

Lesson: *structured task instructions narrow the gap when phrase-type cues are otherwise ambiguous.*

Discussion & Takeaways

Three lessons from operation-aligned semantic evaluation:

1. **Phrasal Semantics Requires Multi-Dimensional Evaluation.** A single task or metric cannot capture phrase-level competence; extraction, classification, and interpretation probe complementary aspects.
2. **Metric Sensitivity Shapes Apparent Strengths.** Strong B-S on interpretation primarily reflects paraphrase fluency; strict span metrics expose brittleness in structural identification. *High interpretation scores* $\not\Rightarrow$ *semantic correctness.*
3. **Workflow Robustness Remains Limited.** Sequential pipelines are highly sensitive to upstream errors; interpretation does not compensate for failed extraction. Cascade fragility stays hidden under flexible metrics.

Conclusion

- **SemanticQA:** an *operation-aligned* benchmark that decomposes phrasal semantics into classification, extraction, and interpretation across IE, LC, NC, and VMwE.
- **Diagnostic, not saturated:** frontier LMs (GPT-5, OpenAI o3, Claude-Sonnet-4.5, DeepSeek-R1, and Kimi-K2) all leave headroom; no model dominates every operation.
- **Cascade-aware:** sequential compositions reveal extraction as the dominant bottleneck for end-to-end semantic pipelines.

Thank You !

🌐 **Project Page:** https://semanticqa.github.io

Code & Data: https://github.com/jacklanda/SemanticQA

🤗 **Datasets:** https://huggingface.co/datasets/jacklanda/SemanticQA